Common Average Reference (CAR)

EEG Preprocessing Method — Summary & Validation

1) What CAR does (short & clear)

Common Average Reference (CAR) is a re-referencing method used in EEG preprocessing. At each time sample, it computes the average signal across all EEG channels and subtracts it from every channel. This reduces common-mode noise shared across electrodes (e.g., reference drift or global interference) while preserving channel-specific brain activity patterns.

2) Why CAR is useful in EEG

EEG channels measure voltage differences relative to a reference. If the reference introduces a shared offset or drift, many channels may shift together. CAR helps remove this shared component, improving the signal-to-noise ratio and making spatial differences between channels clearer for downstream analysis (features, ICA, classification).

3) How we verified CAR works correctly

We validated CAR using both quantitative checks (numbers) and qualitative checks (visual inspection).

Validation Check	Expected Result	Observed Result
Mean across channels after CAR	≈ 0	3.274e-17
Mean across channels before CAR	Not necessarily 0	3.3004
CAR identity check (max abs diff)	Near 0	7.105e-15
CAR identity check (RMSE)	Near 0	3.445e-16

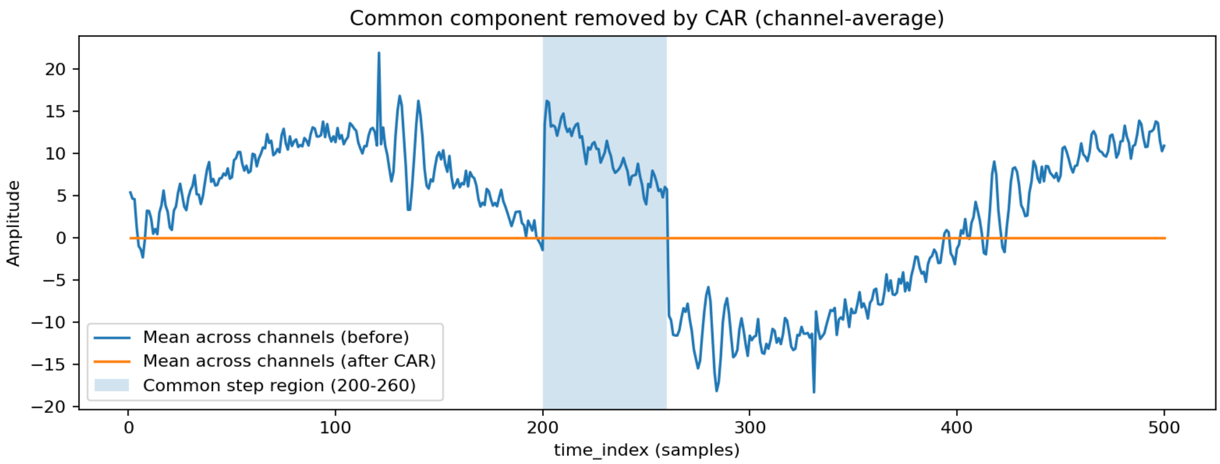

Figure 1 — Channel-average trace before vs after CAR. The mean across channels becomes ~0 after CAR, indicating the common component was removed.

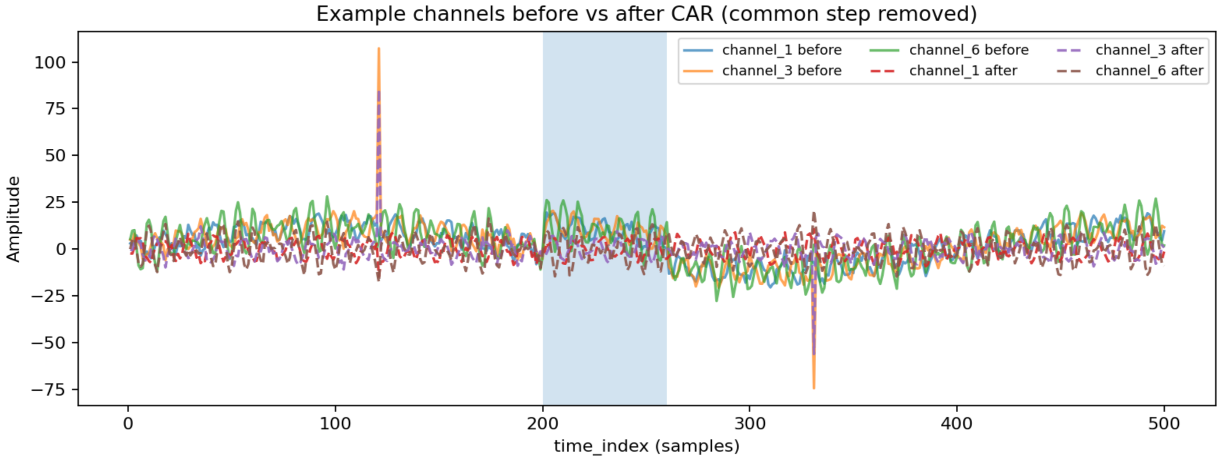

Figure 2 — Example channels before vs after CAR. The shared step artifact (samples 200–260) is reduced after CAR while overall EEG patterns remain.

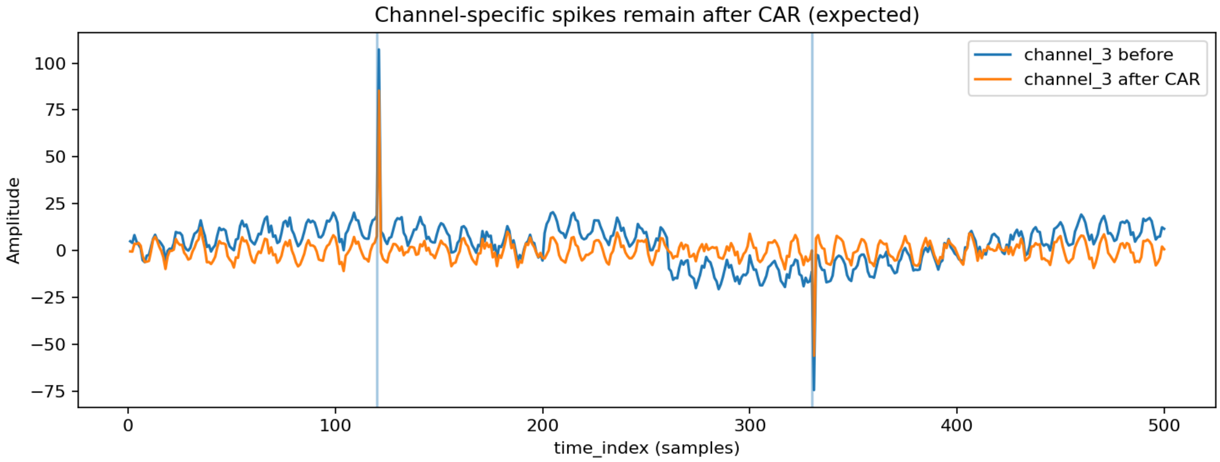

Figure 3 — Channel-specific spikes in channel_3 remain after CAR (expected). CAR removes common-mode noise, not local artifacts.

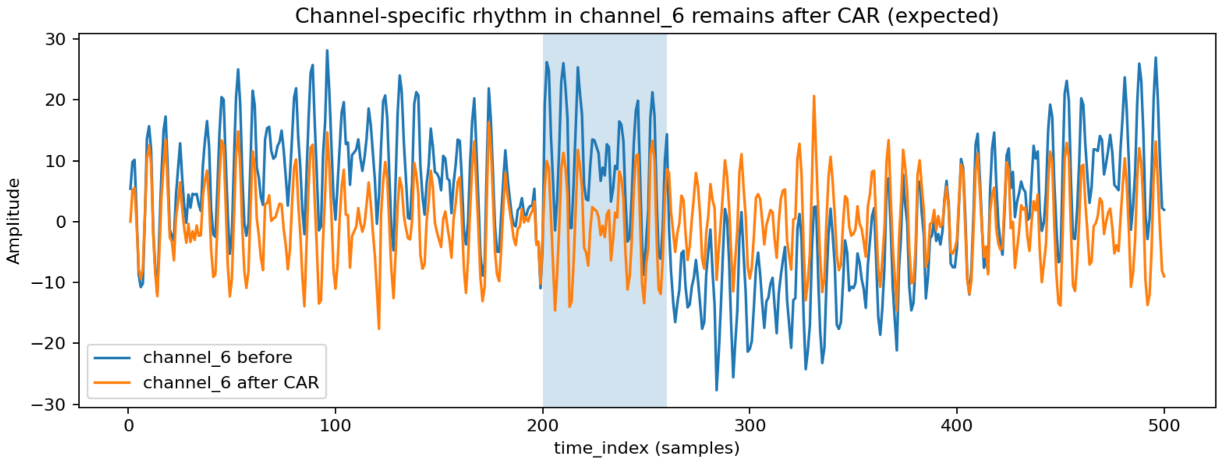

Figure 4 — Channel-specific rhythm in channel_6 remains after CAR (expected), showing CAR preserves non-common neural activity patterns.

Key takeaway

CAR is correct when the channel-average becomes approximately zero after referencing, the dataset structure remains unchanged, and visual inspection shows reduced shared noise while channel-specific activity remains.